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Course: Data Handling and Visualization

SET 26 – STUDENT MINI DATA ANALYSIS

Experiment 6(a): Histogram of Math Score and Boxplot of Science Score by Gender

Aim

To import the Student_Mini_Data.csv dataset in R, create a histogram for Math_Score and a boxplot of Science_Score grouped by Gender, and interpret the results.

Dataset

Student ID	Gender	Study Hours	Attendance	Math Score	Science Score	Exam Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

Procedure

1. Import Student_Mini_Data.csv into R.
2. Check the structure of the dataset.
3. Create a histogram of Math_Score.
4. Create a boxplot of Science_Score grouped by Gender.
5. Add suitable titles and axis labels.
6. Observe the distribution and compare Science scores between genders.

R Code

```
# Import dataset student <-  
read.csv("Student_Mini_Data.csv",  
stringsAsFactors = FALSE)
```

```
# Display dataset print(student)

# Check structure str(student)

# Histogram of Math Score
hist(student$Math_Score, col =
"lightblue", main = "Distribution of
Math Scores", xlab = "Math Score",
ylab = "Frequency")

# Boxplot of Science Score by Gender boxplot(Science_Score
~ Gender,
      data = student, col = c("pink",
"lightgreen"), main = "Science
Score by Gender", xlab =
"Gender", ylab = "Science Score")
```

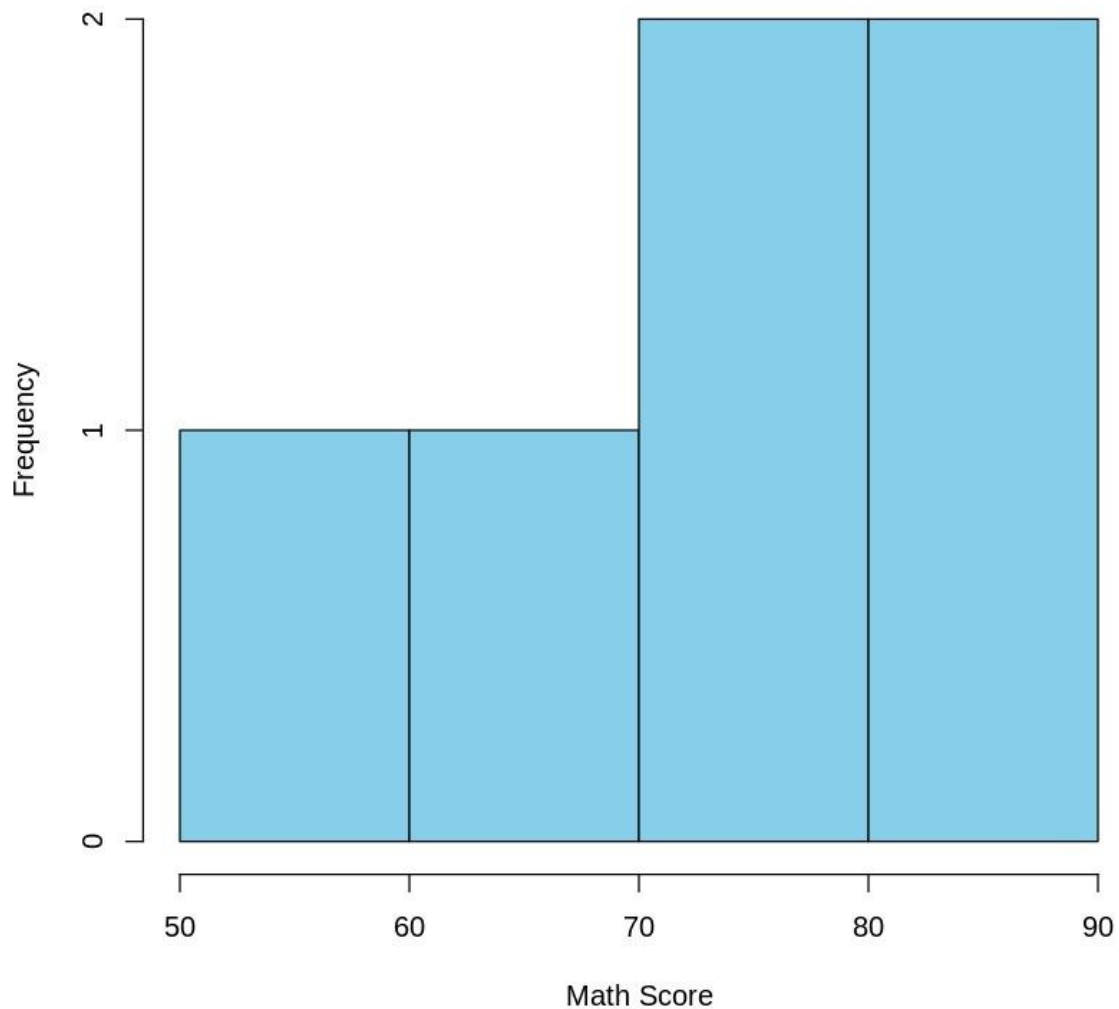
Result / Output

The histogram shows that the Math scores range from **55 to 90**. Female students generally have higher Math scores than male students in this dataset.

The boxplot shows that female students have a higher median Science score compared with male students.

Output: A histogram of Math scores and a gender-wise boxplot of Science scores.

Distribution of Math Scores



Experiment 6(b): Study Hours vs Math Score

Aim

To analyze the relationship between Study_Hours and Math_Score using a scatter plot, with different colors for Gender and a regression line.

Procedure

1. Import the student dataset.
2. Plot Study_Hours on the X-axis.
3. Plot Math_Score on the Y-axis.
4. Use different colors for Male and Female students.
5. Add a regression line.

6. Calculate the correlation between study hours and Math score.
7. Interpret the relationship.

R Code

```
# Assign colors according to gender student$GenderColor
<- ifelse(student$Gender == "Male",
           "blue",
           "red")
```

```
# Scatter plot plot(student$Study_Hours,
student$Math_Score,    pch = 19,
col = student$GenderColor,    main =
"Study Hours vs Math Score",    xlab =
"Study Hours",    ylab = "Math Score")
```

```
# Add regression line abline(lm(Math_Score
~ Study_Hours,
    data = student),
col = "darkgreen",
lwd = 2)
```

```
# Add legend legend("topleft",
legend = c("Male", "Female"),
col = c("blue", "red"),    pch =
19)
```

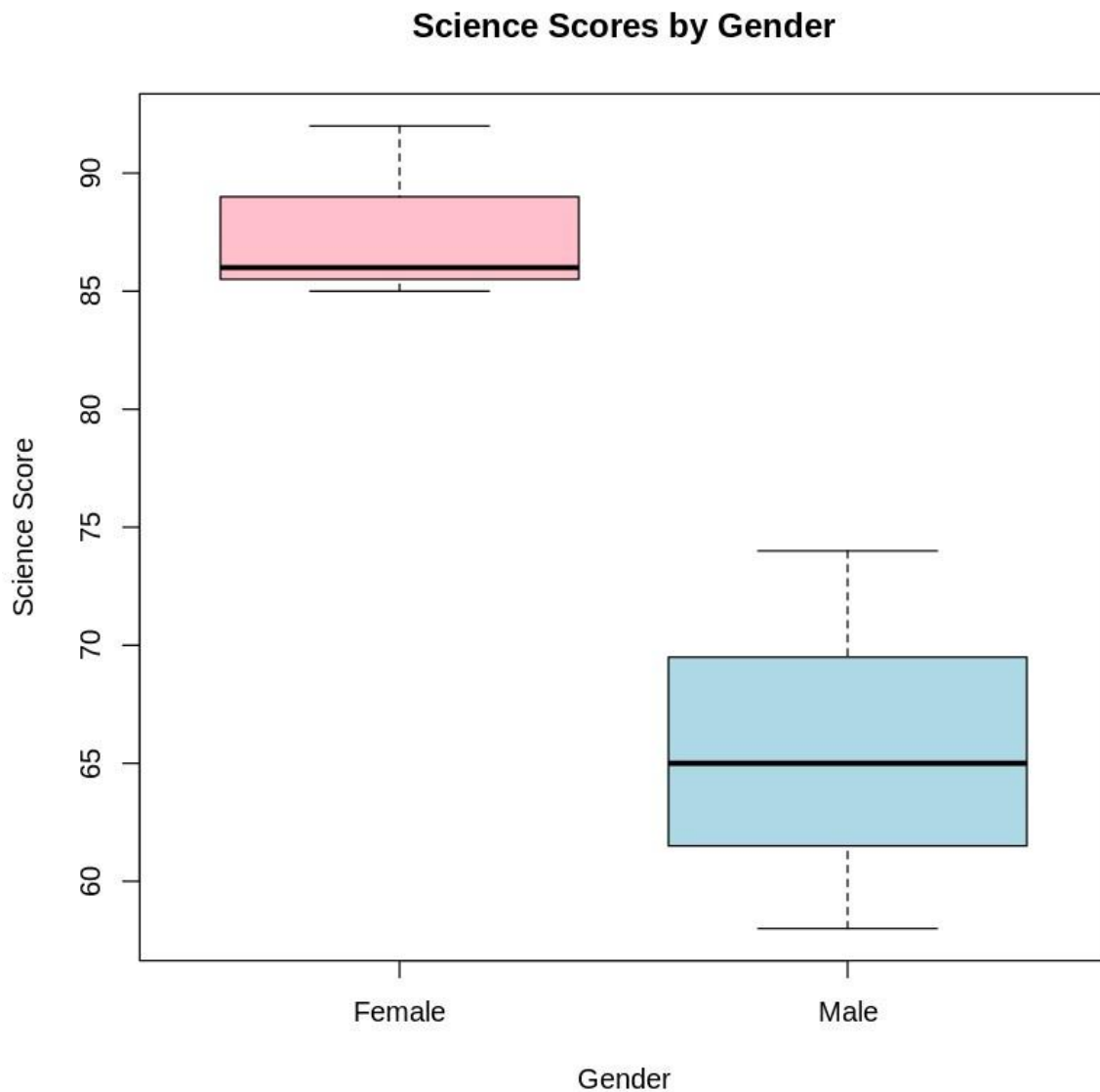
```
# Calculate correlation
cor(student$Study_Hours,
student$Math_Score)
```

Result / Output

The scatter plot shows a **strong positive relationship** between study hours and Math score. Students who spend more hours studying generally obtain higher Math scores.

The regression line slopes upward, indicating that increased study time is associated with improved performance.

Output: A colored scatter plot with a regression line showing the relationship between study hours and Math score.



Experiment 6(c): Monthly Average Math Score with Moving Average

Aim

To convert Exam_Date into Date format, calculate monthly average Math scores, plot the trend, and apply moving-average smoothing.

Procedure

1. Import the student dataset.
2. Convert Exam_Date into Date format.
3. Extract the month from the examination date.
4. Calculate the average Math score for each month.
5. Plot the monthly average scores.
6. Calculate a moving average.
7. Add the moving average to the graph.
8. Identify increasing or decreasing patterns.

R Code

```
# Convert Exam_Date to Date format student$Exam_Date
<- as.Date(student$Exam_Date)

# Extract month student$Month <-
format(student$Exam_Date, "%Y-%m")

# Calculate monthly average Math Score monthly
<- aggregate(Math_Score ~ Month,
              data = student,
              FUN = mean)

# Convert Month to Date monthly$MonthDate
<- as.Date( paste0(monthly$Month, "-01")
)

# Sort by date monthly <-
monthly[order(monthly$MonthDate), ]

# Display monthly averages print(monthly)
```

```
# Plot monthly average Math Score
plot(monthly$MonthDate,
monthly$Math_Score,
      type = "o",    pch = 16,    col =
"blue",    main = "Monthly Average Math
Score",    xlab = "Month",    ylab =
"Average Math Score")

# Calculate 2-month moving average moving_avg
<- stats::filter( monthly$Math_Score,
  rep(1/2, 2),
sides = 1
)
```

```
# Add moving average
lines(monthly$MonthDate,
moving_avg,    type =
"o",    col = "red",
lwd = 2)
```

```
# Add legend legend("topleft",    legend = c("Monthly
Average", "Moving Average"),    col = c("blue",
"red"),    lty = 1,    pch = 16)
```

Result / Output

The monthly average Math scores are:

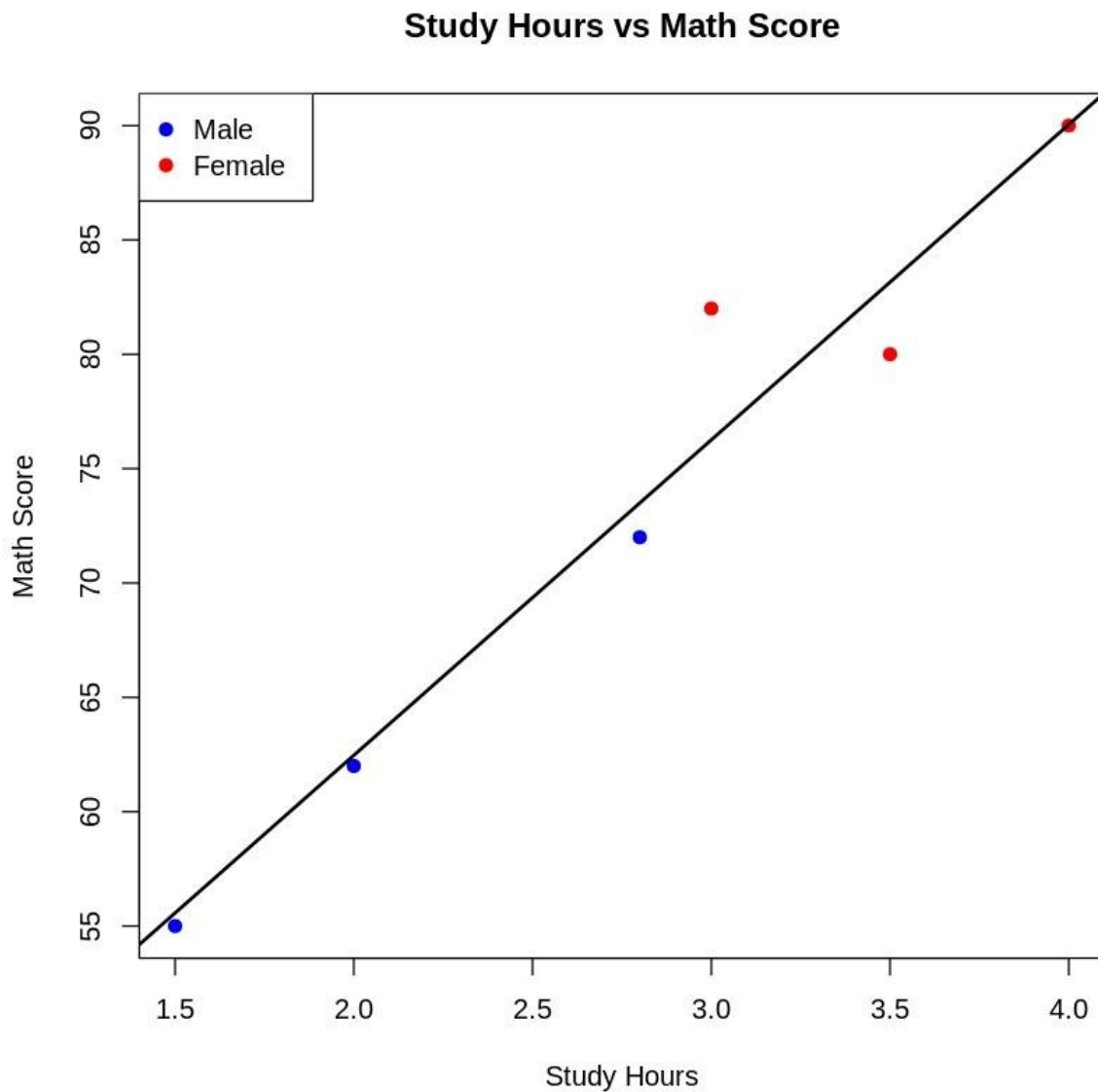
Month	Average Math Score
--------------	---------------------------

January 2025	71
--------------	----

February 2025	72.5
---------------	------

The trend shows an **overall increase in average Math scores** from January to March.

Output: A line chart showing monthly average Math scores and a moving-average line.



Experiment 6(d): Tableau Student Performance Dashboard

Aim

To create an interactive Tableau dashboard showing average Math scores by Gender and score trends over time, with an Exam_Date filter.

Procedure

1. Open **Tableau Desktop**.
2. Select **Connect** → **Text File**.

3. Import Student_Mini_Data.csv.
4. Create a bar chart showing average Math score by Gender.
5. Create a line chart showing Math score trends over Exam Date.
6. Add Exam_Date as a filter.
7. Combine both charts into a dashboard.
8. Add titles, labels and tooltips.
9. Test the filter interaction.

R Code

No R code is required for this question because the visualization is created in Tableau.

Tableau Procedure

Bar Chart – Average Math Score by Gender

Columns → Gender

Rows → AVG(Math_Score)

Marks → Bar

Title:

Average Math Score by Gender

Line Chart – Score Trend

Columns → Exam_Date

Rows → AVG(Math_Score)

Marks → Line

Title:

Math Score Trend Over Time

Add Filter Drag:

Exam_Date

to the **Filters** shelf.

Then select:

Apply to Worksheets → All Using This Data Source

Dashboard

1. Select **New Dashboard**.

2. Add the Gender bar chart.
3. Add the Exam Date line chart.
4. Add the Exam_Date filter.
5. Add appropriate titles and legends.
6. Configure interactive filtering.
7. Save the dashboard.

Result / Output

The Tableau dashboard successfully displays the **average Math score by Gender** and the **Math score trend over examination dates**. The Exam Date filter allows users to interactively view scores for selected dates.

Output: An interactive Tableau dashboard containing a bar chart, line chart and Exam Date filter.

SET 27 – PATIENT HEALTH RISK ANALYSIS

Experiment 7(a): Scatterplot Matrix, Q-Q Plot and ECDF

Aim

To analyze pairwise relationships among **Age, BMI, BP, and Cholesterol** using a scatterplot matrix and assess the distribution of Cholesterol using a Q-Q plot and ECDF.

Dataset

Patient ID Age BMI BP Cholesterol

P1 25 22 120 180

P2 40 28 135 210

P3 55 30 145 240

Patient ID Age BMI BP Cholesterol

P4 35 26 130 200

P5 60 32 150 260

Procedure

1. Create or import the patient dataset in R.
2. Select Age, BMI, BP and Cholesterol.
3. Create a scatterplot matrix using pairs().
4. Create a Q-Q plot for Cholesterol.
5. Add a reference line to the Q-Q plot.
6. Create an ECDF plot for Cholesterol.
7. Examine correlations, distribution and possible skewness.

R Code

```
# Create patient dataset patient
```

```
<- data.frame(  
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),  
  Age = c(25, 40, 55, 35, 60),  
  BMI = c(22, 28, 30, 26, 32),  
  BP = c(120, 135, 145, 130, 150),  
  Cholesterol = c(180, 210, 240, 200, 260)  
)
```

```
# Display dataset print(patient)
```

```
# Scatterplot matrix pairs(patient[,  
  c("Age",  
    "BMI",  
    "BP",  
    "Cholesterol")],  
  main = "Scatterplot Matrix of Health Indicators")
```

```

# Q-Q plot for Cholesterol
qqnorm(patient$Cholesterol,      main
= "Q-Q Plot of Cholesterol")

# Add reference line
qqline(patient$Cholesterol,
col = "red",      lwd = 2)

# ECDF plot
F <- ecdf(patient$Cholesterol)

plot(F,      main = "ECDF of Cholesterol",
xlab = "Cholesterol",      ylab = "Empirical
Cumulative Probability")

```

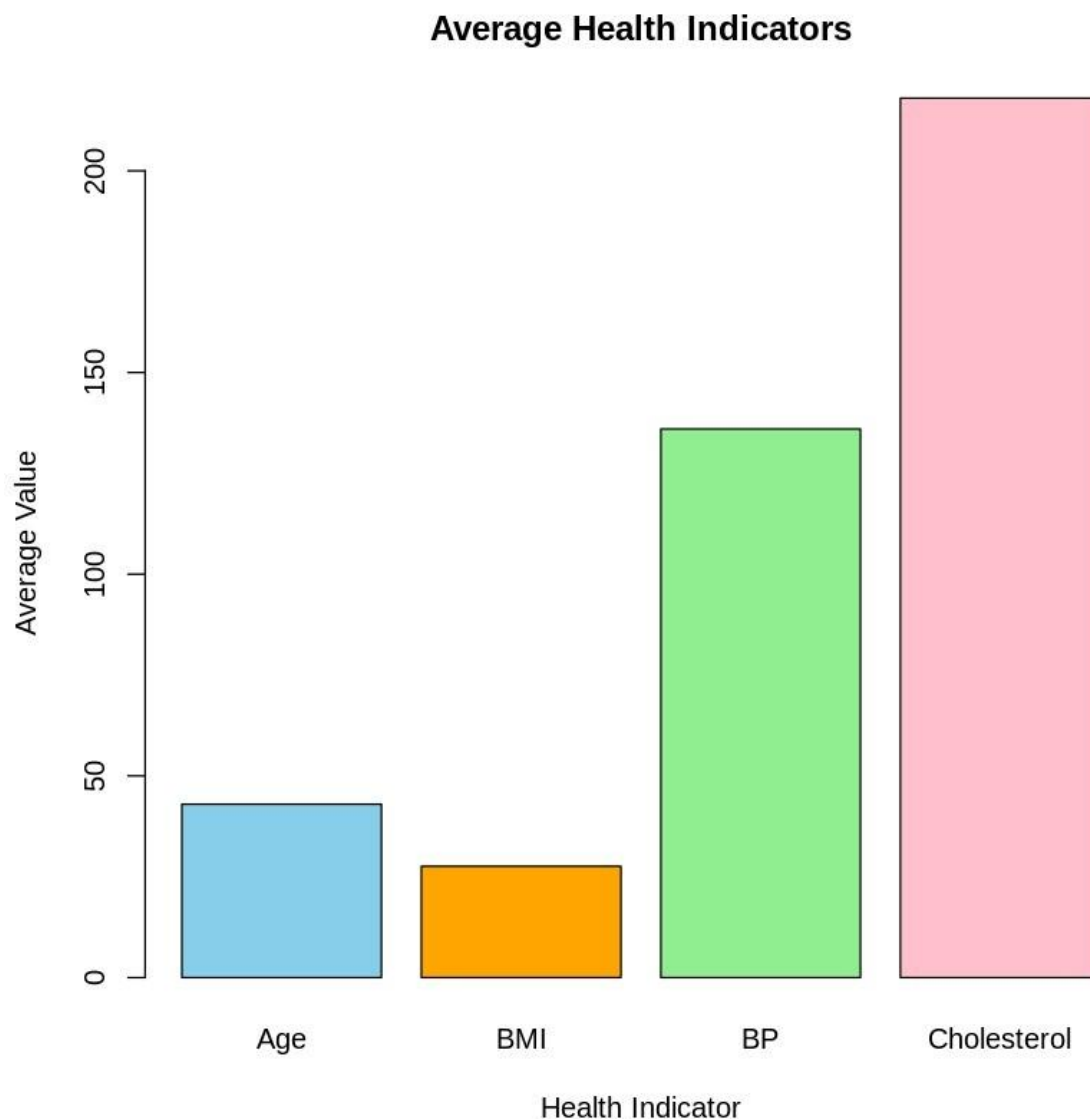
Result / Output

The scatterplot matrix shows that **Age, BMI, BP and Cholesterol generally increase together**, indicating positive relationships among these variables.

The Q-Q plot can be used to assess whether Cholesterol approximately follows a normal distribution. Since there are only five observations, the normality conclusion is limited.

The ECDF shows the cumulative distribution of Cholesterol values from **180 to 260**.

Output: Scatterplot matrix, Q-Q plot and ECDF plot.



Experiment 7(b): Average Health Indicators Bar Chart

Aim

To calculate and visualize the average values of Age, BMI, BP and Cholesterol using a bar chart.

Procedure

1. Create the patient dataset in R.
2. Select the numerical health indicators.
3. Calculate their average values using `colMeans()`.
4. Create a bar chart using the average values.

5. Use different colors for each health indicator.
6. Add value labels.
7. Interpret the averages in relation to general health risk.

R Code

```
# Calculate average values
avg_health <- colMeans(
patient[, c("Age",
           "BMI",
           "BP",
           "Cholesterol")]
)

# Display average values print(avg_health)

# Create bar chart
barplot(avg_health,
col = c("skyblue",
"orange",
       "seagreen",
       "plum"),    main = "Average Patient
Health Indicators", xlab = "Health
Indicator",    ylab = "Average Value",
ylim = c(0, max(avg_health) + 20))

# Add value labels
text(seq_along(avg_health),
avg_health,

      labels = round(avg_health, 1),
pos = 3)
```

Result / Output

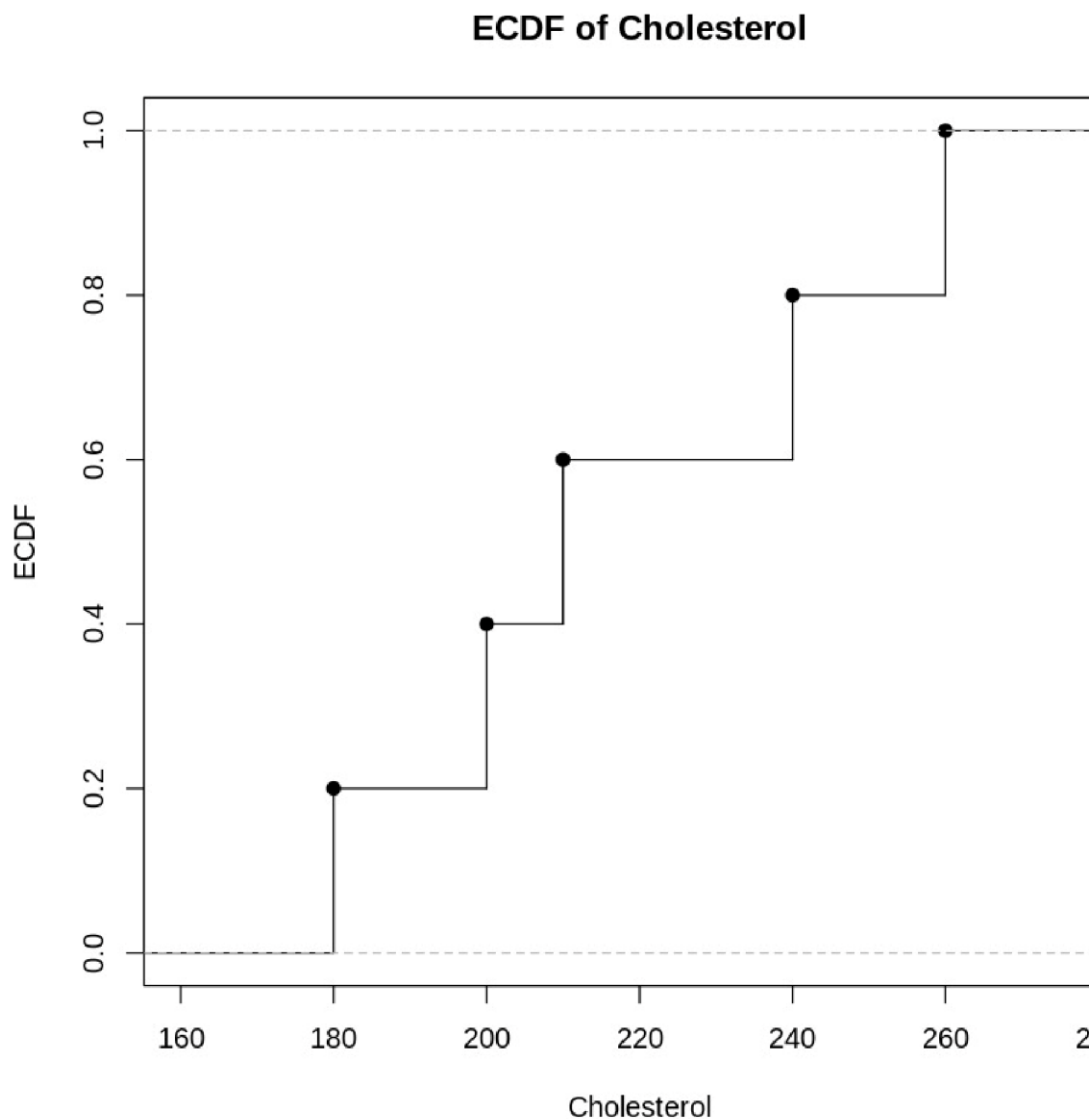
The average values are:

Health Indicator	Average
Age	43
BMI 27.6 BP 136	
Cholesterol	218

Health Indicator

Averag

e



The bar chart summarizes the overall health indicators of the five patients. Higher BMI, blood pressure and cholesterol values can generally be associated with increased health risk, but these values **should not be used alone to diagnose a patient**.

Output: A colored bar chart displaying the average Age, BMI, BP and Cholesterol values.

Experiment 7(c): Interactive Tableau Patient Health Risk Dashboard

Aim

To create an interactive Tableau dashboard with a heatmap for identifying high-risk patient patterns and an Age filter for exploring health indicators.

Procedure

1. Open Tableau Desktop.
2. Import the Patient Health Risk dataset.
3. Create a heatmap using Patient ID and health indicators.
4. Use color intensity to represent higher or lower health indicator values.
5. Add Age as a filter.
6. Add all health indicators to the tooltip.
7. Create a dashboard containing the heatmap.
8. Test the Age filter and tooltip interaction.
9. Analyze which patients have comparatively higher values.

R Code

No R code is required for this question because the visualization is created in Tableau.

Tableau Procedure

Heatmap

Create a new worksheet and use:

Rows → Patient_ID

Columns → Measure Names

Color → Measure Values

Select the following measures:

Age

BMI

BP

Cholesterol

Set the Marks type to:

Square

Use color intensity to identify relatively higher values.

Age Filter

Drag:

Age

to the **Filters** shelf.

Choose the required age range or individual ages.

Tooltip

Add the following fields to the Tooltip:

Patient ID

Age

BMI

BP

Cholesterol

Dashboard

1. Select **New Dashboard**.
2. Drag the heatmap into the dashboard.
3. Add the Age filter.
4. Configure tooltips.
5. Add a descriptive dashboard title:

Patient Health Risk Analysis Dashboard

6. Format the dashboard for easy interpretation.

Result / Output

The Tableau dashboard provides an interactive method for comparing health indicators across patients.

In the supplied data, **P5** has the highest values for Age, BMI, BP and Cholesterol, while **P3** also has comparatively high values. Therefore, these patients show higher values across several indicators and may warrant closer review in an appropriate healthcare setting.

The Age filter allows users to explore how health indicators vary across different age groups.

Output: An interactive Tableau heatmap dashboard with an **Age filter** and detailed **healthindicator tooltips**.



SET 28 – VEHICLE PERFORMANCE ANALYSIS

Experiment 8(a): Violin Plot of Fuel Efficiency by Safety Rating

Aim

To create a violin plot in R to compare the distribution of Fuel_Efficiency across different Safety_Rating categories and analyze the spread, central tendency and variability.

Dataset

Vehicle	Engine Size	Horsepower	Fuel Efficiency	Top Speed	Safety					
ID	(L)	(km/l)	(km/h)	Rating	V1	1.5	110	18	180	4
V2	2.0	150	15	200	5					
V3	3.0	250	12	250	5					
V4	2.5	200	14	220	4					
V5	1.8	130	17	190	3					

Procedure

1. Create the vehicle performance dataset in R.
2. Convert Safety_Rating into a categorical variable.
3. Install and load the ggplot2 package.
4. Create a violin plot with Safety Rating on the X-axis.
5. Place Fuel Efficiency on the Y-axis.
6. Add individual observations using jitter.
7. Analyze the distribution and variability.

R Code

```
# Install package if required
# install.packages("ggplot2")

library(ggplot2)

# Create vehicle dataset vehicle
<- data.frame(
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
```

```

Fuel_Efficiency = c(18, 15, 12, 14, 17),
Top_Speed = c(180, 200, 250, 220, 190),
Safety_Rating = c(4, 5, 5, 4, 3)
)

# Convert Safety Rating to factor vehicle$Safety_Rating
<- as.factor(vehicle$Safety_Rating)

# Violin plot ggplot(vehicle,
aes(x = Safety_Rating, y =
Fuel_Efficiency)) +
geom_violin(fill = "lightblue") +
geom_jitter(width = 0.08,
size = 3, color = "red") +
labs(title = "Fuel Efficiency by Safety
Rating", x = "Safety Rating", y = "Fuel
Efficiency (km/l)"
) +
theme_minimal()

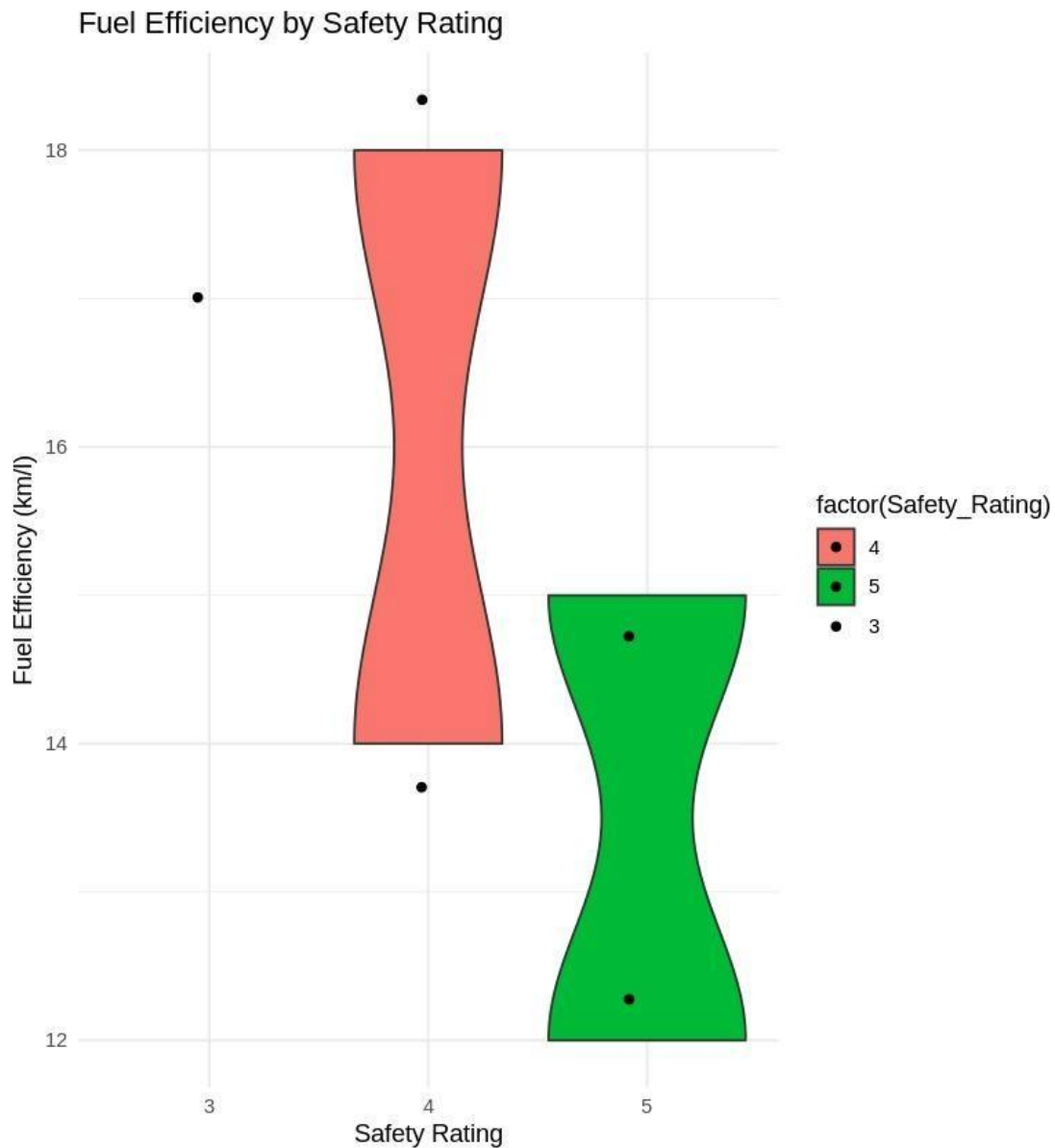
```

Result / Output

The violin plot shows the distribution of fuel efficiency across safety-rating categories. In this small dataset, vehicles with **lower safety ratings tend to have higher fuel efficiency**, while vehicles with a safety rating of 5 have lower fuel efficiency.

Because there are only five vehicles, the distribution should be interpreted cautiously.

Output: A violin plot comparing Fuel Efficiency across Safety Rating categories.



Experiment 8(b): Horsepower vs Top Speed Scatter Plot

Aim

To create a scatter plot showing the relationship between Horsepower and Top_Speed, using different colors based on Engine_Size.

Procedure

1. Create the vehicle dataset in R.
2. Place Horsepower on the X-axis.
3. Place Top Speed on the Y-axis.
4. Use Engine Size to differentiate the observations using color.
5. Add a regression line.

6. Calculate the correlation between Horsepower and Top Speed.
7. Analyze the direction and strength of the relationship. **R Code**

```
library(ggplot2)
```

```
# Scatter plot ggplot(vehicle,  
aes(x = Horsepower, y =  
Top_Speed, color =  
Engine_Size, size =  
Engine_Size)) +  
geom_point(alpha = 0.7) +  
geom_smooth(method = "lm",  
se = FALSE, color  
= "black") + labs( title =  
"Horsepower vs Top Speed", x =  
"Horsepower", y = "Top Speed  
(km/h)", color = "Engine Size (L)",  
size = "Engine Size (L)"  
) +  
theme_minimal()
```

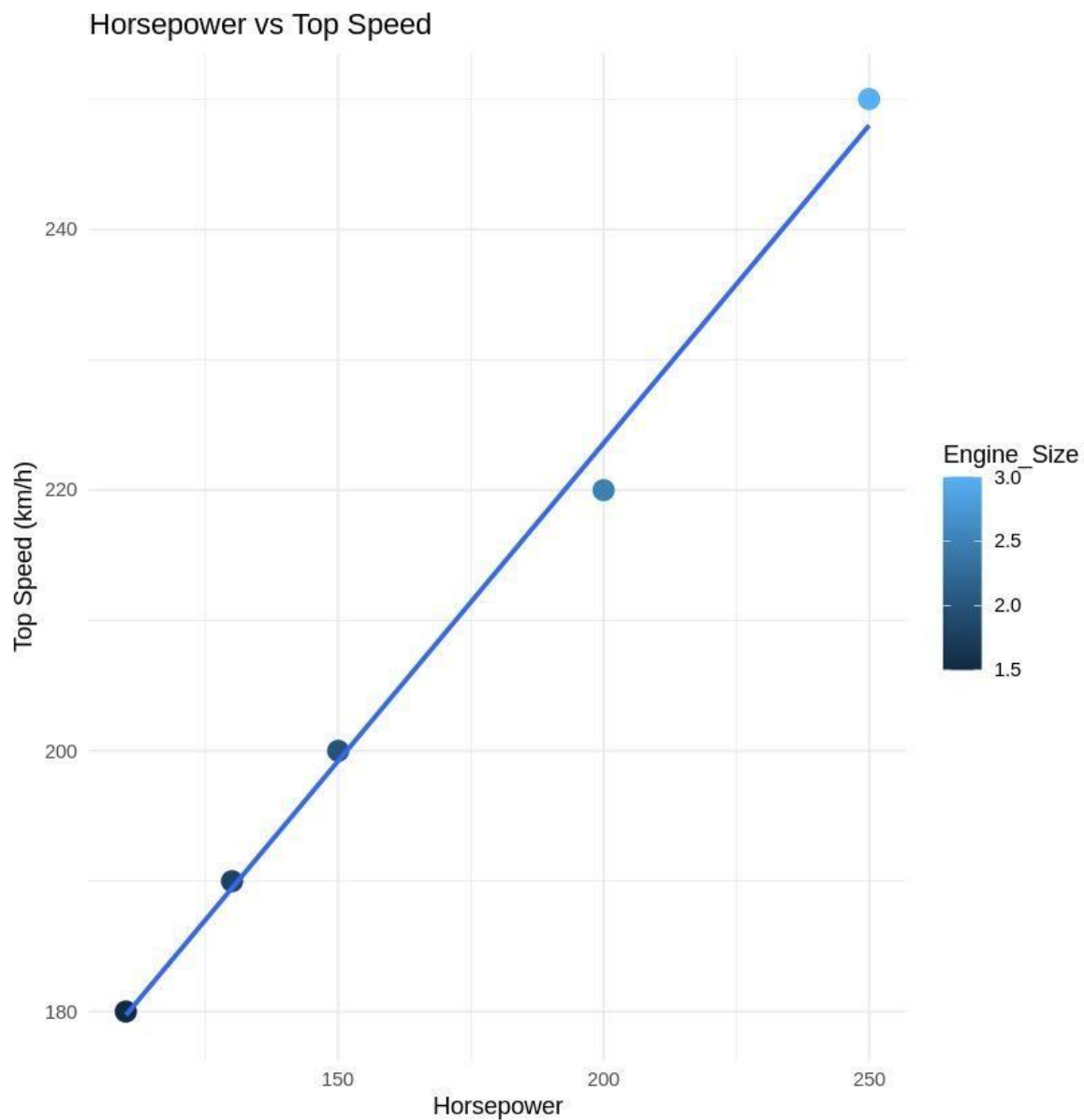
```
# Calculate correlation  
cor(vehicle$Horsepower,  
vehicle$Top_Speed) Result /
```

Output

The scatter plot shows a **strong positive relationship** between Horsepower and Top Speed. Vehicles with higher horsepower generally have higher top speeds.

For this dataset, the correlation is approximately **0.99**, indicating a very strong positive relationship.

Output: A colored scatter plot with a regression line showing the relationship between Horsepower and Top Speed.



Experiment 8(c): Correlation Heatmap of Vehicle Variables

Aim

To create a correlation heatmap for all numerical variables and identify the variables most strongly associated with Top_Speed.

Procedure

1. Select all numerical variables from the vehicle dataset.
2. Calculate the correlation matrix.
3. Install and load the corrplot package.
4. Display the correlation matrix as a heatmap.
5. Identify the variables strongly correlated with Top Speed.

6. Interpret their influence on vehicle performance.

R Code

```
# Install package if required
# install.packages("corrplot")

library(corrplot)

# Select numerical variables numeric_data
<- vehicle[, c(
  "Engine_Size",
  "Horsepower",
  "Fuel_Efficiency",
  "Top_Speed",
  "Safety_Rating"
)]

# Calculate correlation matrix cor_matrix
<- cor(numeric_data)

# Display correlation matrix print(round(cor_matrix,
2))

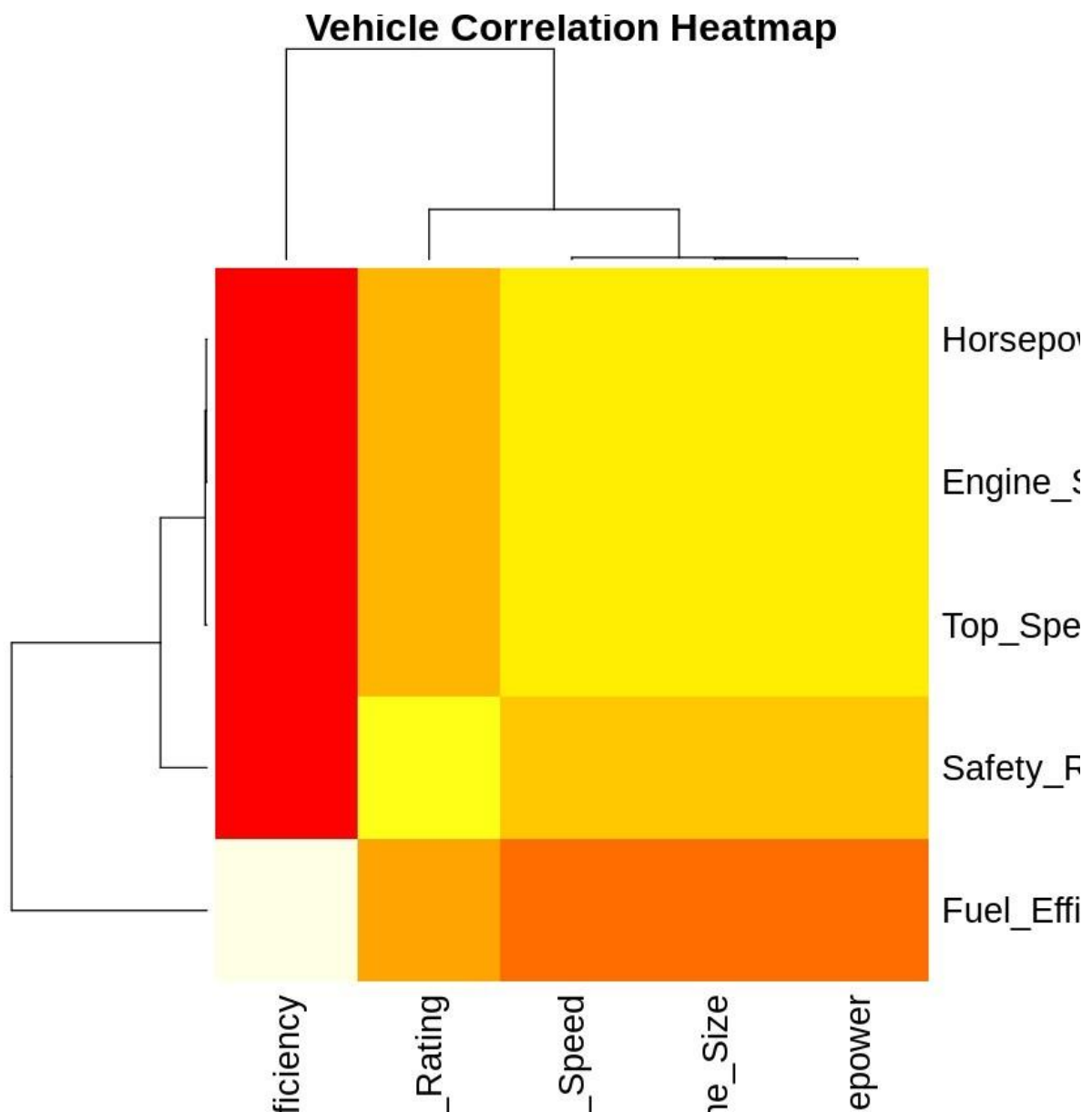
# Create correlation heatmap corrplot(cor_matrix,
  method = "color",      addCoef.col = "black",
  tl.col = "black",      number.cex = 0.8,      title =
"Vehicle Performance Correlation Heatmap",      mar =
c(0, 0, 2, 0))
```

Result / Output

The correlation heatmap indicates that **Horsepower and Engine Size have strong positive relationships with Top Speed**. Fuel Efficiency shows a negative relationship with performance-related variables in this dataset.

Horsepower has the strongest positive association with Top Speed, with a correlation of approximately **0.99**.

Output: A correlation heatmap displaying relationships among the numerical vehicle variables.



Experiment 8(d): Interactive Tableau Vehicle Performance Dashboard

Aim

To design an interactive Tableau dashboard using bar charts and scatter plots with filters for Engine_Size and Safety_Rating to identify vehicles that balance fuel efficiency and high performance.

Procedure

1. Open Tableau Desktop.
2. Import the Vehicle Performance dataset.
3. Create a bar chart comparing Fuel Efficiency across vehicles.
4. Create a scatter plot of Horsepower versus Top Speed.
5. Add Engine Size to color or detail.

6. Add filters for Engine Size and Safety Rating.
7. Combine the charts into a dashboard.
8. Add tooltips showing vehicle details.
9. Analyze vehicles with both good fuel efficiency and high performance. **R**

Code

No R code is required for this question because the visualization is created in Tableau.

Tableau Procedure

Bar Chart – Fuel Efficiency

Columns → Vehicle_ID

Rows → Fuel_Efficiency

Marks → Bar

Title:

Fuel Efficiency by Vehicle

Scatter Plot – Performance

Columns → Horsepower

Rows → Top_Speed

Marks → Circle

Color → Engine_Size

Detail → Vehicle_ID

Title:

Horsepower vs Top Speed

Filters

Add the following fields to the **Filters** shelf:

Engine_Size

Safety_Rating

Apply the filters to all relevant worksheets.

Dashboard

1. Select **New Dashboard**.
2. Add the Fuel Efficiency bar chart.
3. Add the Horsepower vs Top Speed scatter plot.

4. Add Engine Size and Safety Rating filters.
5. Add tooltips containing:
 - Vehicle ID ○
 - Engine Size ○
 - Horsepower ○
 - Fuel
 - Efficiency ○
 - Top Speed ○
 - Safety Rating
6. Add the title **Vehicle Performance Analysis Dashboard**.

Result / Output

The Tableau dashboard provides an interactive comparison of vehicle performance and fuel efficiency.

From the supplied data:

- **V1** has the highest fuel efficiency at **18 km/l**.
- **V3** has the highest top speed at **250 km/h** and highest horsepower at **250 HP**, but its fuel efficiency is the lowest at **12 km/l**.
- **V2** provides a balance of high safety rating, good horsepower and moderate fuel efficiency.
- Higher engine size and horsepower are generally associated with higher top speed.

Output: An interactive Tableau dashboard containing a **fuel-efficiency bar chart**, **performance scatter plot**, and **Engine Size/Safety Rating filters**.

SET 29 – STUDENT ACADEMIC PERFORMANCE

Experiment 9(a): Stacked Area Chart of Test and Participation Scores

Aim

To create a stacked area chart in R showing Test_Score and Participation_Score across students and analyze the performance trends.

Dataset

Student ID	Age	Study Hours	Attendance (%)	Test Score	Participation Score
------------	-----	-------------	----------------	------------	---------------------

S1	19	12	90	85	8
----	----	----	----	----	---

S2	21	8	70	70	7
----	----	---	----	----	---

S3	20	15	95	92	9
----	----	----	----	----	---

S4	22	10	85	80	8
----	----	----	----	----	---

Student ID	Age	Study Hours	Attendance (%)	Test Score	Participation Score
------------	-----	-------------	----------------	------------	---------------------

S5	23	7	60	65	6
----	----	---	----	----	---

Procedure

1. Create the Student Academic Performance dataset in R.
2. Select Test_Score and Participation_Score.
3. Arrange the students in Student ID order.

4. Create a stacked area chart.
5. Add suitable colors for Test Score and Participation Score.
6. Add a title and axis labels.
7. Observe the changes in scores across students.

R Code

```
# Create student dataset student
```

```
<- data.frame(  
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),  
  Age = c(19, 21, 20, 22, 23),  
  Study_Hours = c(12, 8, 15, 10, 7),  
  Attendance = c(90, 70, 95, 85, 60),  
  Test_Score = c(85, 70, 92, 80, 65),  
  Participation_Score = c(8, 7, 9, 8, 6)  
)
```

```
# Create matrix score_data
```

```
<- rbind(  
  student$Test_Score,  
  student$Participation_Score  
)
```

```
# Stacked area chart x
```

```
<- 1:ncol(score_data) #
```

```
Calculate cumulative
```

```
values lower <- c(0,
```

```
score_data[1, ]) upper
```

```
<- score_data[1, ] +
```

```
score_data[2, ]
```

```

plot(x, score_data[1, ], type = "n",
ylim = c(0, max(upper)), xaxt = "n",
main = "Test and Participation Scores",
xlab = "Student", ylab = "Score")

# Test score area polygon(c(x, rev(x)),
c(rep(0, length(x)), rev(score_data[1, ])),
col = "lightblue")

# Participation score area
polygon(c(x, rev(x)),
c(score_data[1, ], rev(upper)),
col = "orange")

# X-axis labels axis(1, at = x,
labels = student$Student_ID) #
Legend legend("topleft",
legend = c("Test Score",
"Participation Score"), fill
= c("lightblue", "orange"))

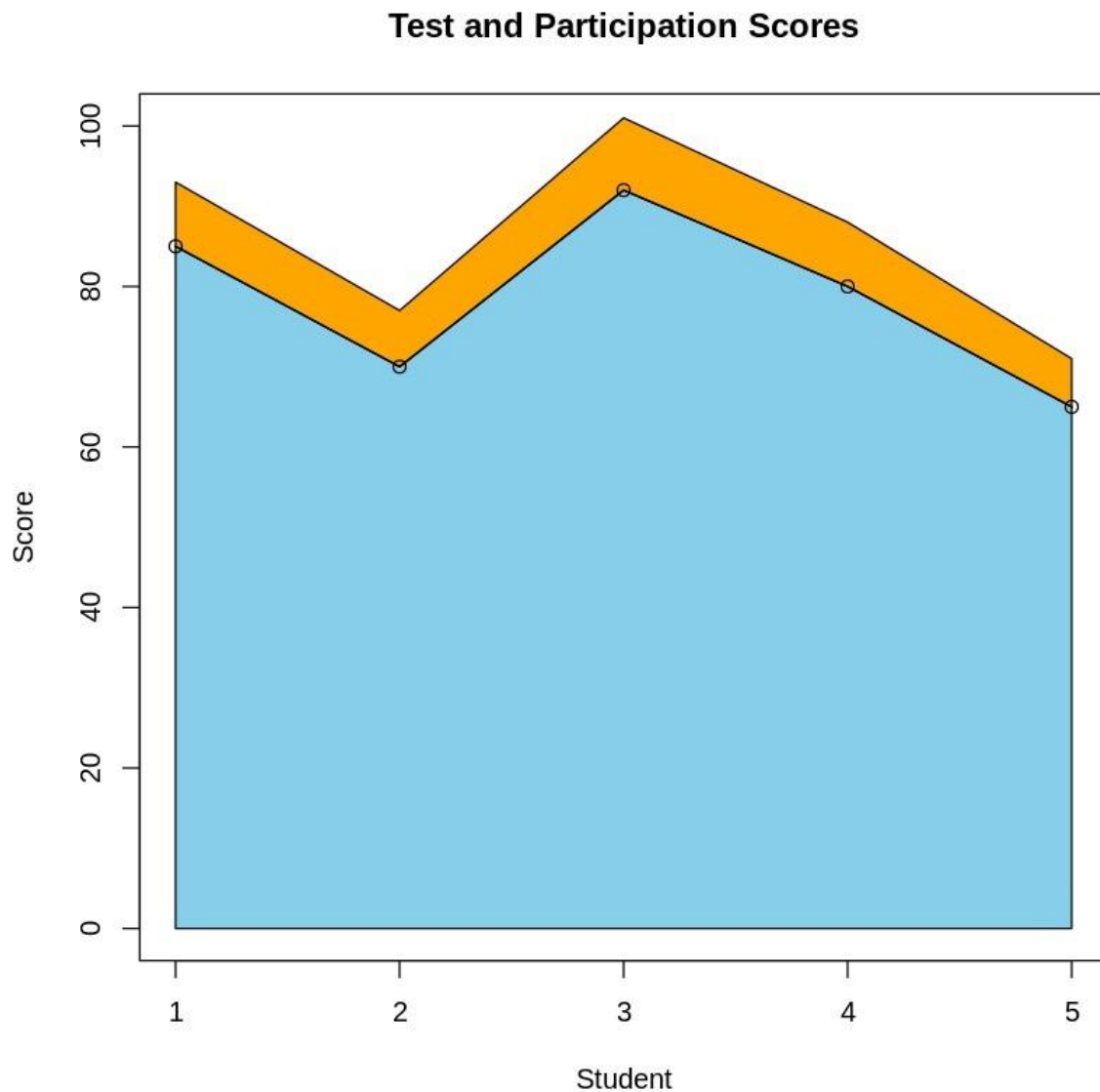
```

Result / Output

The stacked area chart shows variation in Test Score and Participation Score across students. **S3 has the highest Test Score and Participation Score, while S5 has the lowest values.**

Overall, students with stronger academic performance also tend to have higher participation scores in this small dataset.

Output: A stacked area chart displaying Test Score and Participation Score for each student.



Experiment 9(b): Boxplot of Study Hours Grouped by Attendance Quartiles

Aim

To create a boxplot of Study_Hours grouped by attendance quartiles and analyze the median, interquartile range and possible outliers.

Procedure

1. Create the student dataset in R.
2. Calculate the 25th, 50th and 75th percentiles of Attendance.
3. Divide students into attendance quartile groups.
4. Create a boxplot of Study Hours for each attendance group.
5. Add different colors to the groups.

6. Analyze the median, spread and possible outliers.

R Code

```
# Calculate attendance quartiles q
<- quantile(student$Attendance,
             probs = c(0, 0.25, 0.5, 0.75, 1))
```

```
print(q)
```

```
# Create attendance quartile groups
student$Attendance_Quartile <- cut(
  student$Attendance, breaks = q,
  include.lowest = TRUE, labels =
  c("Q1", "Q2", "Q3", "Q4")
)
```

```
# Display groups print(student[,
  c("Student_ID",
    "Attendance",
    "Study_Hours",
    "Attendance_Quartile")])
```

```
# Boxplot boxplot(Study_Hours ~
  Attendance_Quartile,
  data = student,
  col = c("lightblue",
    "lightgreen",
    "orange",
    "pink"),
```

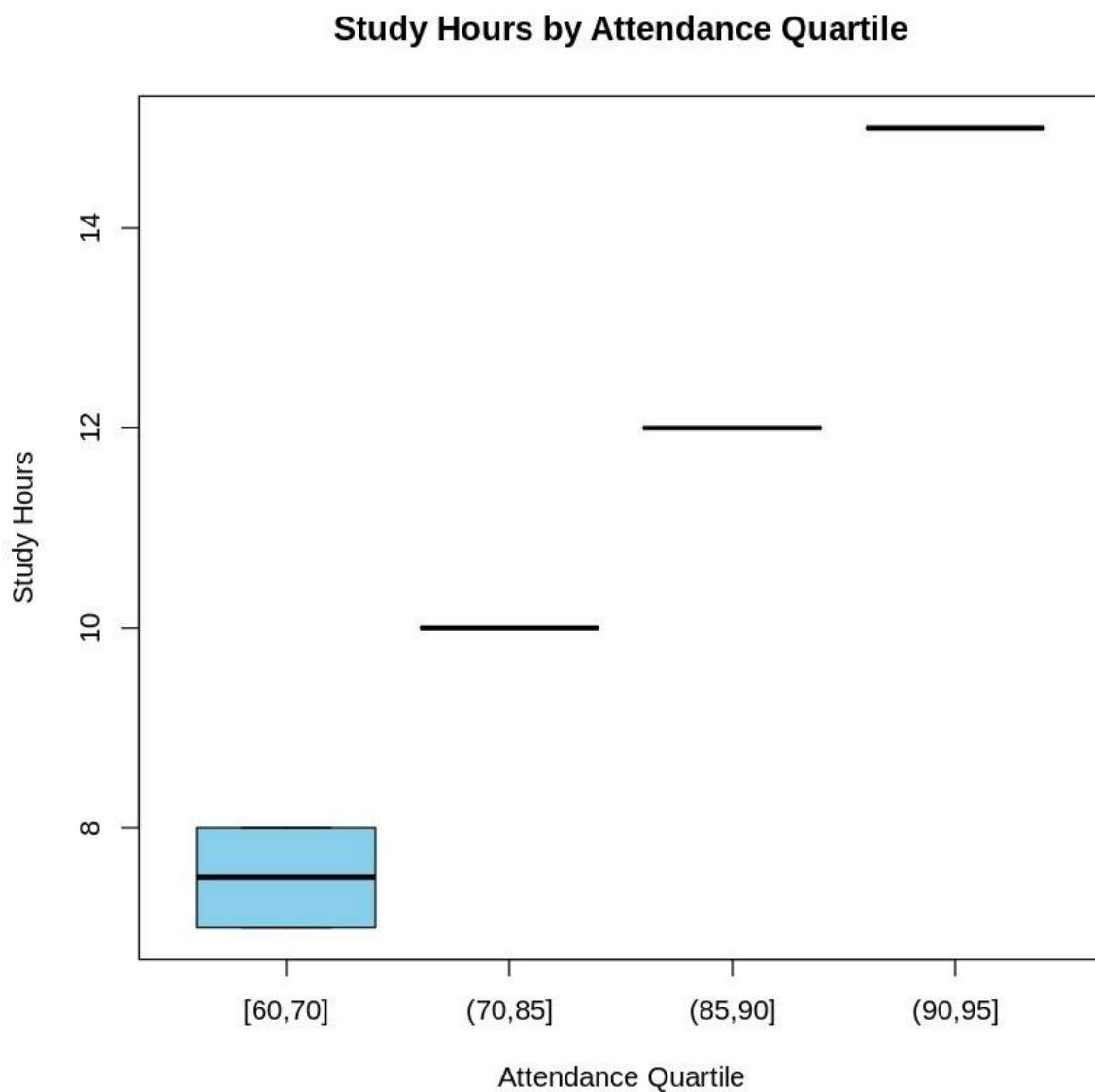
```
main = "Study Hours by Attendance Quartile",  
xlab = "Attendance Quartile",    ylab = "Study  
Hours")
```

Result / Output

The boxplot compares study hours across attendance quartiles. Students with higher attendance generally show higher study hours in this dataset.

However, because there are only **five observations**, some attendance quartile groups contain very few or no observations. Therefore, the median and IQR estimates should be interpreted carefully.

Output: A colored boxplot showing Study Hours grouped by Attendance Quartiles.



Experiment 9(c): Density Plot of Test Scores

Aim

To create a density plot for Test_Score in R to evaluate the score distribution and identify possible unusual values or outliers.

Procedure

1. Create the Student Academic Performance dataset.
2. Select the Test_Score variable.
3. Calculate its density using the density() function.
4. Plot the density curve.
5. Add a suitable title and axis labels.
6. Examine the shape and spread of the distribution.

R Code

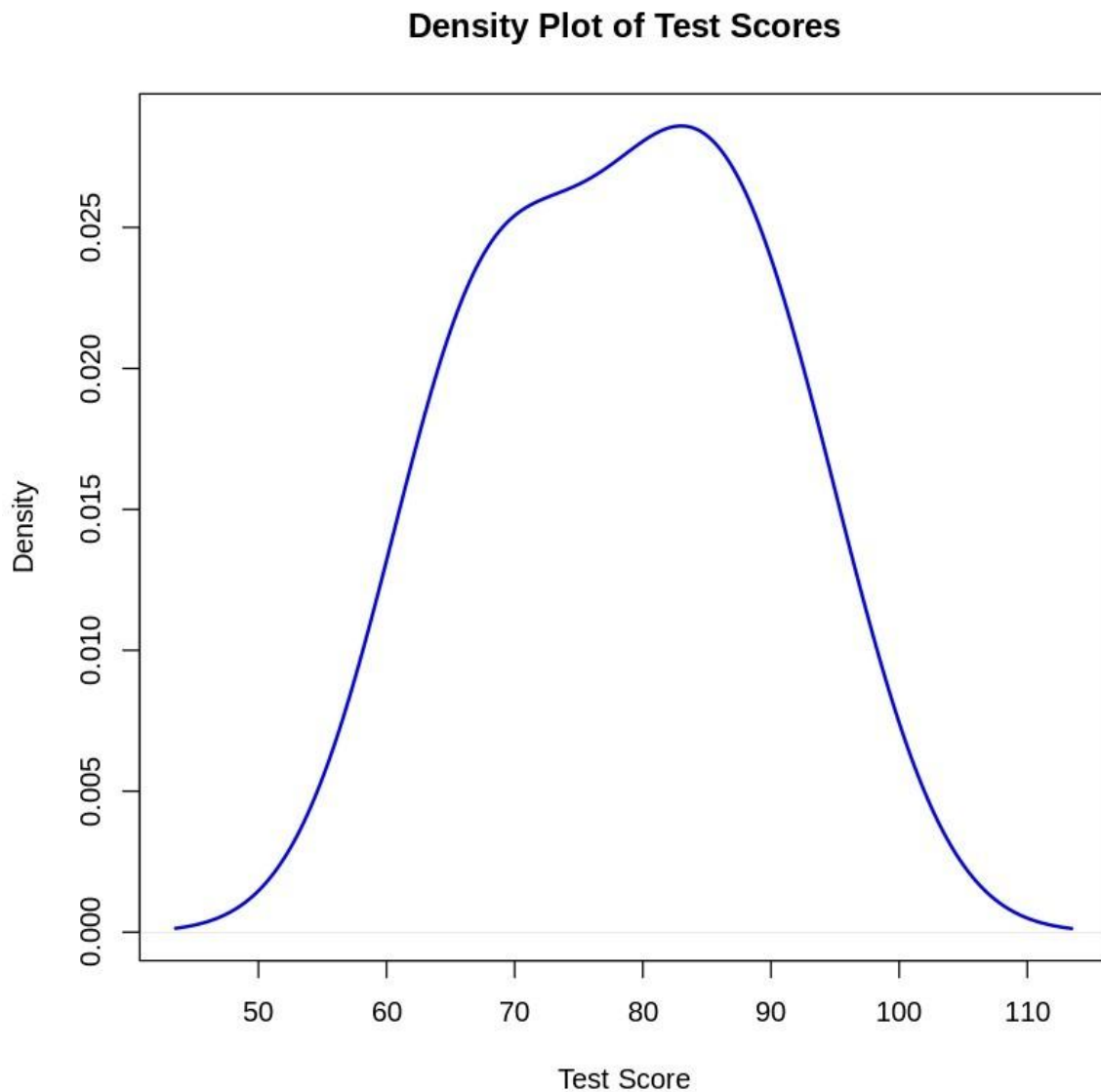
```
# Calculate density test_density <-  
density(student$Test_Score)  
  
# Density plot plot(test_density, col  
= "blue", lwd = 2, main =  
"Density Plot of Test Scores", xlab =  
"Test Score", ylab = "Density")  
  
# Add rug showing individual observations  
rug(student$Test_Score, col = "red")
```

Result / Output

The density plot shows the distribution of Test Scores between **65 and 92**. The data are concentrated around the middle-to-high score range.

There are no obvious extreme outliers in the five observations, although the small sample size limits the reliability of the density estimate.

Output: A density curve showing the distribution of Test Scores with individual observations marked along the X-axis.



Experiment 9(d): Tableau Interactive Student Performance Scatter Plot

Aim

To create an interactive Tableau scatter plot with Study_Hours on the X-axis and Test_Score on the Y-axis, using Participation_Score for color and an Age filter.

Procedure

1. Open Tableau Desktop.
2. Import the Student Academic Performance dataset.
3. Create a scatter plot.
4. Place Study Hours on the X-axis.
5. Place Test Score on the Y-axis.

6. Use Participation Score for color.
7. Add Student ID to Detail.
8. Add Age as a filter.
9. Add tooltips containing student information.
10. Format the visualization and make it interactive.

R Code

No R code is required for this question because the visualization is created in Tableau.

Tableau Procedure

Scatter Plot

Columns → Study_Hours

Rows → Test_Score

Marks → Circle

Color → Participation_Score

Detail → Student_ID

Chart title:

Study Hours vs Test Score

Age Filter

Drag: Age

to the **Filters** shelf.

Choose the required age values or range.

Tooltip Add:

Student ID

Age

Study Hours

Attendance

Test Score

Participation Score to

the Tooltip.

Dashboard

1. Select **New Dashboard**.
2. Add the scatter plot.
3. Add the Age filter.
4. Enable tooltips.
5. Format the chart.
6. Add the title **Student Academic Performance Analysis**.

Result / Output

The Tableau scatter plot shows the relationship between **Study Hours and Test Score**. The supplied data indicate that students who study more hours generally achieve higher test scores. Color intensity represents Participation Score, allowing students with different participation levels to be compared.

For example, **S3** has the highest study hours (15), highest test score (92), and highest participation score (9).

Output: An interactive Tableau scatter plot with **Study Hours, Test Score, Participation Score color coding**, and an **Age filter**.

SET 30 – MOBILE APP USAGE ANALYSIS

Experiment 10(a): Histogram and Density Plot of Screen Time

Aim

To import the Mobile_App_Usage.csv dataset in R and create a histogram and density plot for Screen_Time to analyze the distribution of mobile screen usage.

Dataset

User	Screen Time	App Usage	Data Used	Usage
-------------	--------------------	------------------	------------------	--------------

Gender Age			Satisfaction		
ID		(hrs)	Count	(GB)	Date
2025-01-					
U01	Male	20 4.5	18	2.4	3
08					
2025-01-					
U02	Female	22 6.0	25	3.8	5
08					
2025-02-					
U03	Male	19 3.2	12	1.6	3
11					
2025-02-					
U04	Female	21 7.1	30	4.5	5
11					
2025-03-					
U05	Male	23 2.8	10	1.2	2
14					
2025-03-					
U06	Female	20 5.4	22	3.1	4
14					

Procedure 1. Import Mobile_App_Usage.csv into R.

2. Display the dataset and check its structure.
3. Select the Screen_Time variable.
4. Create a histogram to show the frequency distribution.
5. Add a density curve to the histogram.
6. Add suitable colors, labels and title.
7. Interpret the distribution of screen time.

R Code

```
# Import dataset mobile <-
read.csv("Mobile_App_Usage.csv",
```



```

stringsAsFactors = FALSE)

# Display dataset print(mobile)

# Check structure str(mobile)

# Histogram of Screen Time
hist(mobile$Screen_Time, probability
= TRUE, breaks = 6, col =
"lightblue", main = "Distribution of
Screen Time", xlab = "Screen Time
(Hours)", ylab = "Density")

# Add density curve lines(density(mobile$Screen_Time),
col = "red",
lwd = 2)

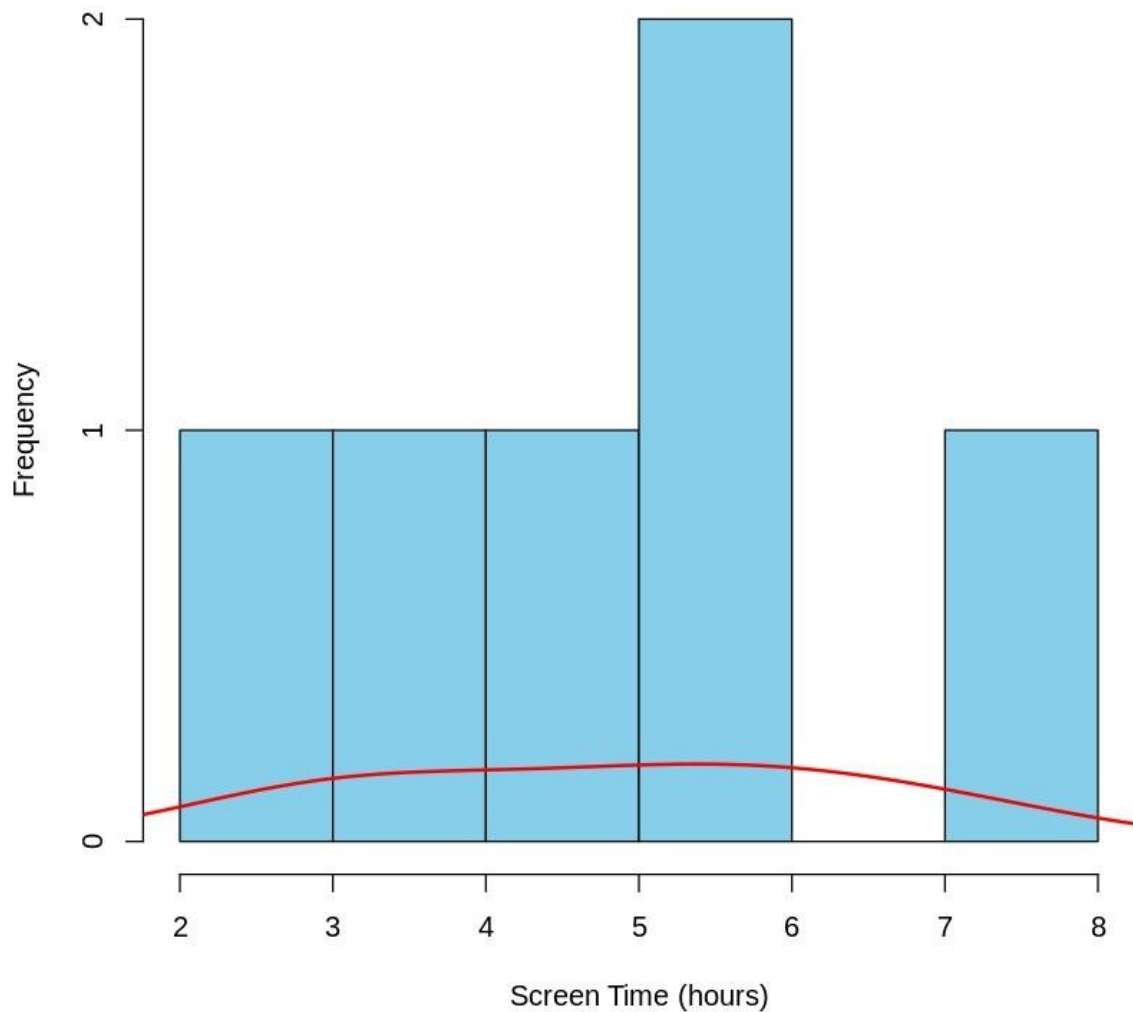
```

Result / Output

The screen-time values range from **2.8 to 7.1 hours**. The histogram and density curve show variation in users' daily screen time. Female users in this dataset generally have higher screen-time values than male users.

Output: A histogram with a density curve showing the distribution of Screen Time.

Distribution of Screen Time



Experiment 10(b): Data Used vs Screen Time Scatter Plot

Aim

To create a scatter plot between Data_Used and Screen_Time, calculate their correlation, add a trend line, and analyze the relationship.

Procedure

1. Import the Mobile App Usage dataset.
2. Place Data Used on the X-axis.
3. Place Screen Time on the Y-axis.
4. Create a scatter plot.
5. Add a regression trend line.

6. Calculate the correlation coefficient.
7. Analyze the relationship between data usage and screen time.

R Code

```
# Scatter plot plot(mobile$Data_Used,
mobile$Screen_Time,    pch = 19,
col = "blue",    main = "Data Used vs
Screen Time",    xlab = "Data Used
(GB)",    ylab = "Screen Time
(Hours)")

# Add trend line abline(lm(Screen_Time
~ Data_Used,
    data = mobile),
col = "red",    lwd
= 2)

# Calculate correlation correlation <-
cor(mobile$Data_Used,
mobile$Screen_Time)

print(correlation)
```

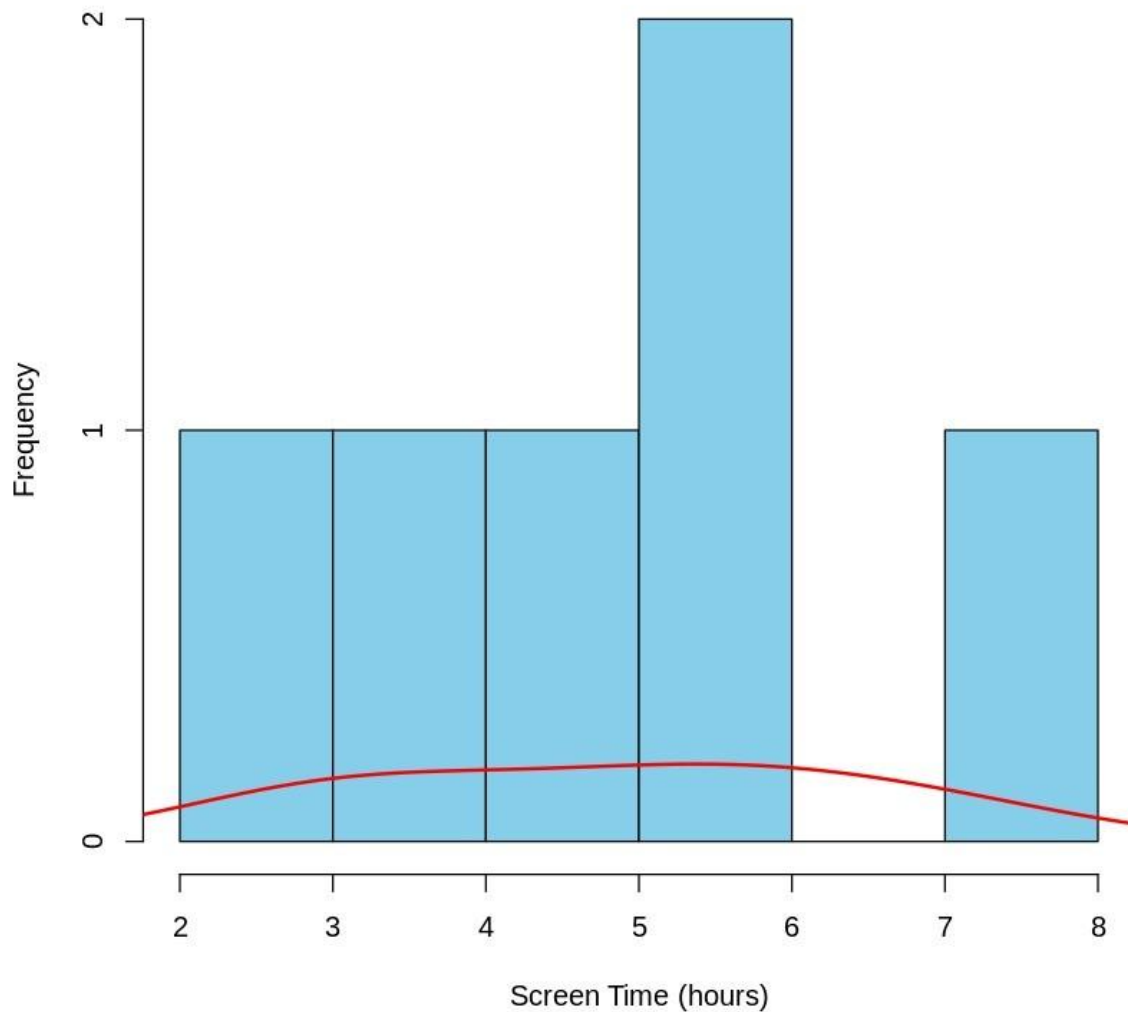
Result / Output

The scatter plot shows a **strong positive relationship** between Data Used and Screen Time. Users with higher screen time generally consume more mobile data.

For this dataset, the correlation is approximately **0.99**, indicating a very strong positive relationship.

Output: A scatter plot with a regression trend line showing the relationship between Data Used and Screen Time.

Distribution of Screen Time



Experiment 10(c): Average Satisfaction by Gender

Aim

To calculate the average Satisfaction score for each gender and create a bar chart with value labels to compare user satisfaction patterns.

Procedure

1. Import the Mobile App Usage dataset.
2. Group the observations by Gender.
3. Calculate the average Satisfaction score for each gender.
4. Create a bar chart of the averages.
5. Add value labels to the bars.

6. Compare satisfaction between male and female users.

R Code

```
# Calculate average satisfaction by gender
avg_satisfaction <- aggregate(
  Satisfaction ~ Gender, data = mobile,
  FUN = mean
)

# Display averages print(avg_satisfaction)

# Create bar chart
barplot(avg_satisfaction$Satisfaction,
  names.arg = avg_satisfaction$Gender, col
  = c("skyblue", "pink"), main = "Average
  Satisfaction by Gender", xlab =
  "Gender", ylab = "Average Satisfaction
  Score", ylim = c(0, 5.5))

# Add value labels
text(seq_along(avg_satisfaction$Satisfaction),
  avg_satisfaction$Satisfaction, labels =
  round(avg_satisfaction$Satisfaction, 2), pos =
  3)
```

Result / Output

The average satisfaction scores are:

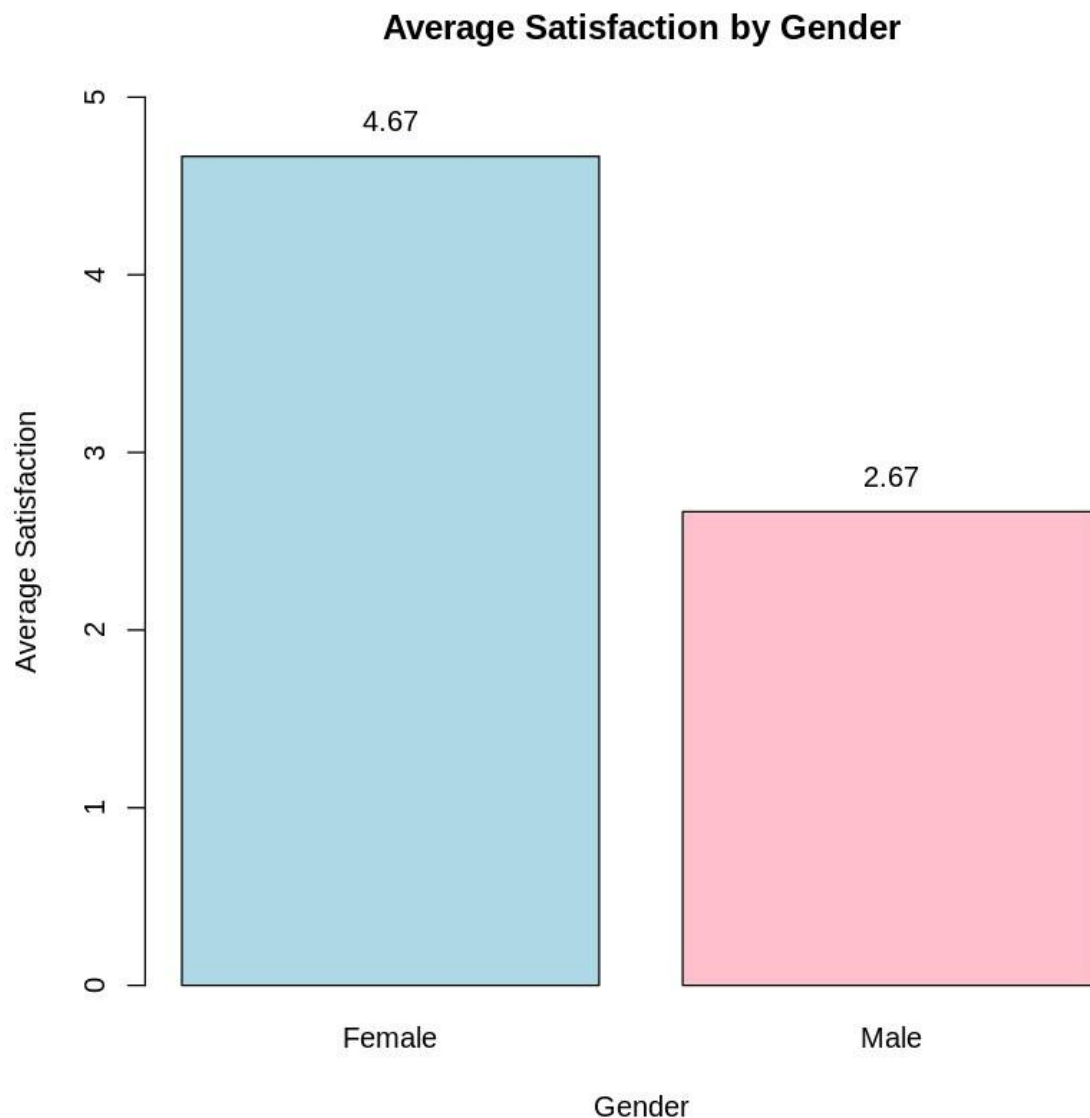
Gender Average Satisfaction

Male 2.67

Female 4.67

Female users have a **higher average satisfaction score** than male users in this dataset.

Output: A bar chart comparing average satisfaction scores by gender with values displayed above the bars.



Experiment 10(d): Tableau Mobile App Usage Mini Dashboard

Aim

To create an interactive Tableau mini dashboard containing a bubble chart of Screen Time vs Data Used and a line chart showing usage trends, with filters for Gender and Usage Date.

Procedure

1. Open Tableau Desktop.
2. Select **Connect** → **Text File**.
3. Import Mobile_App_Usage.csv.
4. Create a bubble chart using Screen Time and Data Used.

5. Use Gender for color.
6. Use App Usage Count for bubble size.
7. Create a line chart for usage trends over Usage Date.
8. Add Gender and Usage Date filters.
9. Combine both visualizations into a dashboard.
10. Add titles, tooltips and formatting.

R Code

No R code is required for this question because the visualization is created in Tableau.

Tableau Procedure

Bubble Chart – Screen Time vs Data Used

Columns → Screen_Time

Rows → Data_Used

Marks → Circle

Color → Gender

Size → App_Usage_Count

Detail → User_ID

Chart title:

Screen Time vs Data Used

Line Chart – Usage Trends

For the line chart, use Usage Date and Screen Time:

Columns → Usage_Date

Rows → AVG(Screen_Time)

Marks → Line

Chart title:

Mobile Usage Trend

You can also use App_Usage_Count instead of Screen Time if your faculty expects appusage-count trends.

Filters

Add the following fields to the Filters shelf:

Gender

Usage_Date

Apply the filters to all worksheets.

Dashboard

1. Select **New Dashboard**.
2. Drag the bubble chart into the dashboard.
3. Drag the line chart into the dashboard.
4. Add the Gender filter.
5. Add the Usage Date filter.
6. Configure the filters to apply to both charts.
7. Add tooltips containing:
 - User ID ○ Gender
 - Screen Time ○ App
 - Usage Count ○ Data
 - Used ○ Satisfaction
8. Add the title:

Mobile App Usage Analysis Dashboard

Result / Output

The Tableau mini dashboard successfully displays the relationship between **Screen Time and Data Used** using a bubble chart. Gender is represented using color and App Usage Count using bubble size.

The line chart displays mobile usage trends over time. The Gender and Usage Date filters allow users to interactively analyze specific groups and dates.

Output: An interactive Tableau mini dashboard containing a **bubble chart, usage trend line chart, Gender filter, and Usage Date filter**.